

MATUKU, Fiji

WMO: 916970

Lat: 19.133S

Lon: 179.750E

Elev: 5

StdP: 101.26

Time Zone: 12.00 (PCF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	19.8	12.0	8.7	21.5	13.0	9.3	21.7	9.5	22.4	8.5	23.4	1.5	160	0.272

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.1	33.0	29.3	31.9	28.4	30.9	27.2	29.6	32.5	28.9	31.7	27.7	30.6	3.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.9	25.5	32.1	28.0	24.2	31.7	26.7	22.3	30.2	97.9	32.4	94.1	31.7	88.2	30.6	31.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.0	6.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.4	27.4	27.7	27.7	26.6	25.3	24.0	23.0	23.1	23.7	24.5	25.8	26.7		
	DBStd	2.22	1.39	1.43	1.64	1.22	1.22	1.23	1.24	1.49	1.67	1.56	1.58	1.56		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	5634	540	494	548	498	474	419	403	407	411	448	474	517		
	CDD18.3	2592	282	261	289	248	216	169	145	148	161	190	224	259		
	CDH23.3	20187	3023	2897	3186	2317	1480	679	336	417	649	1007	1747	2451		
	CDH26.7	4268	851	901	980	406	100	14	9	7	51	106	302	541		
(14) Wind	WSAvg	2.6	2.5	2.3	2.4	2.8	2.6	2.4	2.3	2.7	2.7	2.9	2.8	2.6		
(15) Precipitation	PrecAvg	2113	286	244	268	230	159	117	94	95	114	140	159	208		
	PrecMax	2951	661	425	417	410	360	290	162	247	219	264	406	393		
	PrecMin	1425	55	63	123	68	46	37	24	24	19	35	52	71		
	PrecStd	385	141	91	81	99	81	63	38	53	55	61	80	72		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	33.5	34.1	30.9	28.9	27.7	27.7	27.6	28.7	29.7	31.6	32.5		
		MCWB	29.0	29.4	30.0	26.6	25.1	24.4	23.9	23.6	24.4	25.3	28.2	28.3		
	2%	DB	32.1	32.4	33.3	30.0	28.2	27.1	26.3	26.7	28.0	28.7	30.6	31.7		
		MCWB	28.6	28.7	29.9	25.7	24.5	23.6	22.9	23.2	24.2	24.6	27.2	28.2		
	5%	DB	30.8	31.2	32.0	29.3	27.7	26.5	25.6	26.0	27.1	27.9	29.6	30.7		
		MCWB	27.4	27.5	28.9	25.3	24.0	23.1	22.3	22.7	23.6	23.9	25.8	27.1		
	10%	DB	29.8	30.2	30.1	28.6	27.2	25.9	25.0	25.4	26.2	27.0	28.4	29.1		
		MCWB	26.0	26.3	26.1	24.9	23.6	22.6	21.7	22.0	22.6	23.1	24.4	25.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.7	30.0	30.6	27.5	26.4	25.8	25.3	24.7	25.3	26.0	28.8	29.2		
		MCDB	31.9	32.8	33.3	30.0	27.9	27.0	27.1	26.4	27.7	28.6	31.0	31.7		
	2%	WB	29.1	29.1	30.0	26.7	25.6	24.7	24.0	23.8	24.8	25.1	27.8	28.5		
		MCDB	31.7	31.9	33.1	28.9	27.5	26.2	25.4	26.0	27.2	28.1	30.4	31.5		
	5%	WB	27.9	27.9	29.0	26.1	24.9	23.9	23.2	23.2	24.1	24.3	26.2	27.2		
		MCDB	31.1	31.1	32.4	28.2	26.9	25.7	24.9	25.4	26.6	27.2	28.7	30.3		
	10%	WB	26.5	26.9	26.8	25.6	24.3	23.3	22.4	22.7	23.3	23.8	25.1	25.9		
		MCDB	29.1	29.6	29.4	27.8	26.6	25.4	24.5	24.9	25.7	26.4	27.6	28.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	4.1	3.8	3.4	3.4	3.2	3.3	3.3	3.3	3.4	3.6	3.6		
		MCDBR	4.4	4.7	3.5	4.2	3.8	3.6	4.1	3.5	3.6	3.8	4.2	4.0		
		MCWBR	2.1	2.3	1.8	2.2	2.1	2.1	2.4	2.1	2.2	2.0	2.2	2.0		
	5% WB	MCDBR	3.6	4.2	2.5	4.2	3.4	3.2	3.8	3.4	3.1	3.3	3.6	3.4		
		MCWBR	2.0	2.1	1.4	2.4	2.2	2.2	2.8	2.3	2.1	2.0	2.0	1.9		
(40) Clear-Sky Solar Irradiance	taub		0.384	0.380	0.378	0.377	0.372	0.371	0.357	0.362	0.377	0.381	0.379	0.384		
	taud		2.634	2.653	2.671	2.654	2.629	2.618	2.646	2.637	2.582	2.583	2.619	2.629		
	Ebn at Noon		961	953	933	895	863	845	870	898	919	941	959	961		
	Edh at Noon		101	98	93	89	85	83	82	89	100	104	102	102		
(44) All-Sky Solar Radiation	RadAvg		6.55	6.18	5.57	4.71	4.11	3.63	3.89	4.42	5.20	5.83	6.32	6.47		
	RadStd		0.59	0.46	0.35	0.40	0.29	0.27	0.19	0.44	0.50	0.46	0.52	0.32		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.21	N/A	N/A	N/A	N/A	N/A	+0.25	N/A	N/A	+85

Nomenclature: See separate page